THE STRUCTURE MORPHISM OF PROJECTIVE SPACE IS SEPARATED, QUASI-COMPACT, VALUATIVE AND UNIVERSALLY CLOSED

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Allowed standard prerequisites (editorial).

- Separatedness of Proj. lemma-proj-separated (carded this batch). Part (1) over Spec(Z).
- Standard covering. lemma-standard-covering-projective-space (carded this batch). Part (2): quasi-compactness over Spec(Z).
- Valuative criterion. lemma-proj-valuative-criterion (carded this batch). Part (3) over Spec(Z).
- Base-change permanence. schemes-lemma-separated-permanence, schemes-lemma-quasi-compact-preserved-base-change, schemes-proposition-characterize-universally-closed (cross-chapter, named). Pull the properties back to S and close the universally-closed part.

Source reference. Original lemma 01MC: sources/stacks-constructions/pinned-original-source.tex, source lines 1532–1532. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MF: sources/stacks-constructions/pinned-original-source.tex, source lines 1602–1602. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NG: sources/stacks-constructions/pinned-original-source.tex, source lines 3077–3077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01K5: focused-seeds/geometry/S9-cons-projective-space-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KU: focused-seeds/geometry/S9-cons-projective-space-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KF: focused-seeds/geometry/S9-cons-projective-space-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. The structure morphism $\mathbf{P}^n_S \to S$ is*

(1) *separated,*
(2) *quasi-compact,*
(3) *satisfies the existence and uniqueness parts of the valuative criterion, and*
(4) *universally closed.*

Proof. All these properties are stable under base change (this is clear for the last two and for the other two see Schemes, Lemmas 01KU and 01K5). Hence it suffices to prove them for the morphism $\mathbf{P}^n_{\mathbf{Z}} \to \mathrm{Spec}(\mathbf{Z})$. Separatedness is Lemma 01MC. Quasi-compactness follows from Lemma 01NG. Existence and uniqueness of the valuative criterion follow from Lemma 01MF. Universally closed follows from the above and Schemes, Proposition 01KF. □